

# Perceiver & Perceiver IO

## Implementation and Experimental Analysis

Francesco Gigli and Corso Vignoli

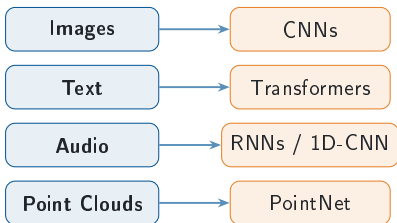
Università degli Studi di Firenze  
Course B031278 — Deep Learning

# Agenda

- 1 Introduction
- 2 Perceiver
- 3 Experiments: Perceiver
- 4 Perceiver IO
- 5 Experiments: Perceiver IO
- 6 Conclusion

# The Problem: Modality-Specific Architectures

## One modality, one architecture



Each data type is usually paired with a dedicated model and domain-specific assumptions about structure.

## The scaling bottleneck

### Self-attention cost

$$\mathcal{O}(M^2 \cdot d)$$

grows quadratically with input length  $M$

| Input representation         | $M$    |
|------------------------------|--------|
| $16 \times 16$ image patches | 256    |
| raw $224 \times 224$ pixels  | 50,176 |

Raw pixels make standard attention  
**impractical.**

**Core question:** can one architecture handle arbitrary inputs without quadratic cost?

# The Perceiver Solution

**Core idea:** learned latents read the input once; expensive processing stays in latent space.

## 1. Read

### Cross-attention

$M$  inputs  $\rightarrow N \ll M$  latents

## 2. Process

### Latent block

transform  $N$  latent slots

## 3. Reuse

### Shared weights

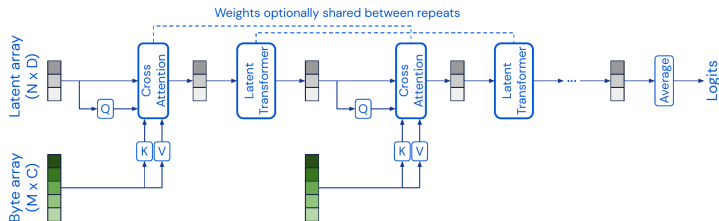
repeat the same processor

## Complexity shift

Read the full input once; repeat the expensive work in the compact latent space  $N \ll M$ .

| Operation                      | Cost                       |
|--------------------------------|----------------------------|
| Read input once                | $\mathcal{O}(M \cdot N)$   |
| Process latents for $L$ layers | $\mathcal{O}(L \cdot N^2)$ |
| <b>Perceiver</b>               | $\mathcal{O}(MN + LN^2)$   |

# Perceiver Architecture



## Encode and process

- $X \in \mathbb{R}^{M \times C}$  supplies keys and values.
- $Z \in \mathbb{R}^{N \times D}$  supplies queries, with  $N \ll M$ .
- Cross-attention maps  $X \rightarrow Z$ :  $\mathcal{O}(MN)$ .

## Latent core and output

- Latent Transformer updates  $Z$  only.
- Repeated blocks may share weights.
- Original Perceiver pools  $Z$ ; Perceiver IO uses output queries.

# Cross-Attention: The Key Mechanism

Formally:

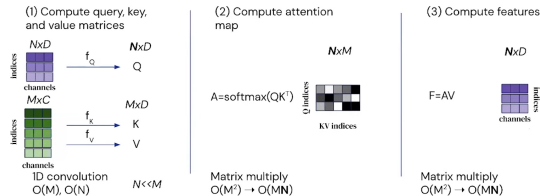
$$Q = Z W_Q \in \mathbb{R}^{N \times d_k}$$

$$K = X W_K \in \mathbb{R}^{M \times d_k}$$

$$V = X W_V \in \mathbb{R}^{M \times d_k}$$

$$\text{CrossAttn}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

The **latents  $Z$  are the queries**, the input  $X$  provides keys and values  $\Rightarrow$  informational *bottleneck*.

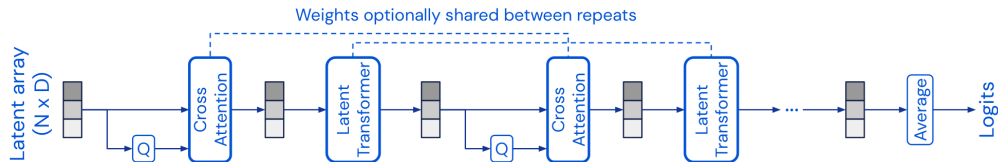


Key insight:

- Attention matrix:  $N \times M$  (not  $M \times M$ )
- Complexity:  $\mathcal{O}(N \cdot M \cdot d_k)$
- Decouples depth from input size

# Latent Transformer & Weight Sharing

Once information is in the latent array, depth is spent by reusing the same processor on a compact state.



## Weight sharing

- The same Latent Transformer block  $B_\theta$  is reused at every repeat, with shared parameters  $\theta$ .
- Unrolling depth behaves like an RNN over the latent state:  $Z_{t+1} = B_\theta(Z_t)$ .

## Why it matters

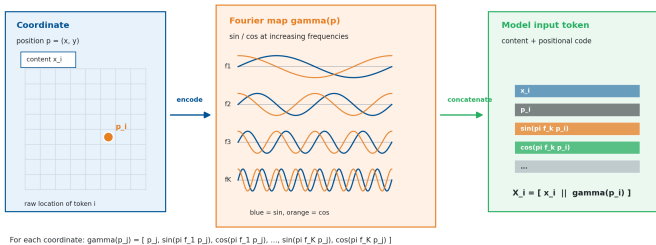
- Iterative refinement without instantiating a new processor at each step.
- Parameter count stays fixed as depth grows; computation stays in the  $N$ -slot latent space.

# Fourier Positional Encoding

Perceiver receives a flat array, so location has to be part of each input token.

## Fourier features turn position into a multiscale signature

The input stays a flat array: every element receives content plus its encoded coordinates.



## General recipe

Use domain coordinates: images  $(u, v)$ , point clouds  $(x, y, z)$ , or text index  $t$ ; sine/cosine bands encode them at multiple scales.

## Input seen by attention

The array is still flat, but each token is enriched before projection:  $X_i = [x_i \parallel \gamma(p_i)]$ .

# Perceiver Forward Pass: Shape Checkpoints

## Data flow in the original Perceiver:

- 1 Input array: each token stacks content  $x$  with a Fourier position code  $\gamma(p)$ ,

$$X = [x \parallel \gamma(p)] \in \mathbb{R}^{M \times C}.$$

- 2 Learned latent array:

$$Z_0 \in \mathbb{R}^{N \times D}, \quad N \ll M$$

- 3 Cross-attention reads input:

$$Q = ZW_Q, \quad K = XW_K, \quad V = XW_V$$

- 4 Latent Transformer applies self-attention + FFN only on  $N$  latents.

## Shape checkpoints:

| Object | Shape          | Meaning                   |
|--------|----------------|---------------------------|
| $X$    | $M \times C$   | input array (content+pos) |
| $Z_0$  | $N \times D$   | learned latents           |
| $Q$    | $N \times d_k$ | latent queries            |
| $K, V$ | $M \times d_k$ | input keys/values         |
| $A$    | $N \times M$   | attention map             |

## Core idea

The input size  $M$  affects only the cross-attention read; depth is spent in the compact latent space.

# Original Perceiver Output: Pooling Classifier

## Fixed-shape output head:

$$Z_T = \text{Encoder}(X, Z_0)$$

$$z = \frac{1}{N} \sum_{i=1}^N Z_T[i]$$

$$\text{logits} = zW_{\text{cls}} + b_{\text{cls}}$$

- Mean pooling collapses all latents to one vector
- The classifier predicts one global label
- This is ideal for classification, not dense structured outputs

## Where FFN and sharing fit:

- Each latent block is pre-norm attention, residual, FFN, residual
- FFN adds non-linearity after attention mixing
- Weight sharing repeats the processor over iterations

## Fixed-shape output

Pooling collapses the  $N$  latents into a single vector, so this head emits one global prediction per input.

# Computational Complexity Analysis

| Model                | Self-Attn            | Cross-Attn         | Total                         |
|----------------------|----------------------|--------------------|-------------------------------|
| Standard Transformer | $\mathcal{O}(M^2 d)$ | —                  | $\mathcal{O}(M^2 d \cdot L)$  |
| Perceiver            | $\mathcal{O}(N^2 d)$ | $\mathcal{O}(MNd)$ | $\mathcal{O}((MN + N^2 L) d)$ |

## Symbols:

- $M$ : number of input elements
- $N$ : number of latent slots, with  $N \ll M$
- $L$ : number of latent processing layers
- $d$ : hidden channel dimension

## Interpretation:

- The full input is read through cross-attention
- Repeated self-attention happens only in latent space
- If  $N$  is fixed, depth becomes independent of input length
- Concrete sizes are evaluated in the experiments

# GELU Activation & LAMB Optimizer

## GELU (Gaussian Error Linear Unit):

$$\text{GELU}(x) = x \cdot \Phi(x) \approx x \cdot \sigma(1.702 x)$$

- Smooth activation used in Transformer architectures
- Allows small negative gradient flow

## LAMB (Layer-wise Adaptive Moments):

$$r_t^{(l)} = \frac{\|w_t^{(l)}\|}{\|w_t^{(l)} + \eta \hat{u}_t^{(l)}\|}$$

Per-layer trust ratio rescales updates and stabilizes large-batch training.

## Why LAMB for Perceiver?

- Cross-attention bottleneck creates high variance gradients
- Standard Adam can **destabilize** with large batches
- LAMB safely scales batch sizes up to 1024

| Optimizer   | Large Batches | Layer-wise LR          |
|-------------|---------------|------------------------|
| Adam        | Diverges      | Global                 |
| <b>LAMB</b> | <b>Stable</b> | <b>Per-layer trust</b> |

# Implementation

| Aspect              | Paper implementation                                                             | Project code                                                                                       |
|---------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Core model          | Cross-attention into learned latents, latent self-attention, FFN, weight sharing | Same modules; weight sharing is exposed as an ablation switch                                      |
| Model scale         | Large vision/language runs: up to $512 \times 1024$ latents and 201M-param IO    | Smaller runs: CIFAR-10 $96 \times 384$ , ModelNet40/text $128 \times 512$                          |
| Data setting        | ImageNet, ModelNet40, large language pretraining corpora                         | CIFAR-10, ModelNet40, WikiText-103 byte-level MLM, GLUE fine-tuning                                |
| Training budget     | Distributed accelerators, batches up to 512–1024                                 | Single-GPU budget, batches 32–64, LAMB retained for stability                                      |
| Practical additions | Minimal regularization in the reference setup                                    | Validation split carved from train, fixed seed, attention-map extraction, declarative run registry |

The implementation keeps the architectural mechanism intact; the main differences are scale, data budget and training resources.

# Implementation Differences from the Paper

| Choice                      | Original paper                                                                                | This project                                                                                                                              |
|-----------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Input tokenization          | Raw pixels, no patching ( $M = 50,176$ )                                                      | <b>Raw pixels, no patching</b> ( $M = 1,024$ on $32 \times 32$ )                                                                          |
| Latent array $N \times D$   | $512 \times 1024$                                                                             | $96 \times 384$ (CIFAR-10), $128 \times 512$ (ModelNet40)                                                                                 |
| Cross-attend iterations $T$ | 8                                                                                             | 4 (CIFAR-10), 2 (ModelNet40) — and $T$ is itself an ablation axis                                                                         |
| Positional encoding         | Fourier, $K = 64$ bands/axis (ImageNet: 258 dims; ModelNet40 too)                             | CIFAR-10 <b>identical</b> : 258 dims + 3 RGB = 261.<br>ModelNet40 and text: $K = 6$ ( $3 + 39 = 42$ per point, $257 + 13 = 270$ per byte) |
| Augmentation, masking       | ModelNet40: scale and shift jittered <i>per point</i> . MLM: 15% of <i>whole words</i> masked | Scale and shift of the <i>whole</i> cloud, undone by the normalisation. MLM: 15% of <i>single bytes</i> — an easier task                  |
| Weight sharing              | On by default (shared across $T$ )                                                            | Exposed as an ablation switch (on/off)                                                                                                    |
| Model selection             | —                                                                                             | <b>Validation carved from train</b> (45k/5k/10k); test touched once. ModelNet40 has no validation split: epoch picked on test             |
| Parameters, MLP width       | 44.9M on ImageNet; MLP $1 \times$ (hidden = $D$ )                                             | 10.18M (CIFAR-10), 23.29M (ModelNet40), 18.87M (IO text); MLP $4 \times$                                                                  |

## Kept identical to the paper

Core mechanism preserved: cross-attention bottleneck  $\rightarrow$  weight-shared latent transformer  $\rightarrow$  mean-pooling (Perceiver) / output-query decoder (IO); GELU MLP (but  $4 \times$  wide, the paper uses  $1 \times$ ), pre-norm LayerNorm, Fourier frequencies linearly spaced to Nyquist.

# CIFAR-10: Experimental Setup

**Goal:** validate the Perceiver on raw pixels, with no convolutional assumption anywhere.

| Hyperparameter                | Value                     |
|-------------------------------|---------------------------|
| Latent array ( $N \times D$ ) | $96 \times 384$           |
| Cross-attn stages ( $T$ )     | 4, interleaved            |
| Transformer blocks ( $L$ )    | 4 (shared)                |
| Heads (cross / self)          | 1 / 8                     |
| Fourier bands / max freq      | 64 / 16                   |
| Dropout                       | 0                         |
| Optimizer                     | LAMB                      |
| Learning rate                 | 0.004                     |
| Scheduler                     | MultiStep 84/102/114      |
| Batch size                    | 64                        |
| Epochs                        | 120, <i>no early stop</i> |
| Precision                     | bfloat16                  |
| <b>Total parameters</b>       | <b>10 175 362</b>         |

## Protocol — the part that makes the numbers usable

Train split: **45 000** images. Validation: **5 000**, carved out of the training set. Test: the official **10 000**, touched *once*, at the very end.

The reported checkpoint is the one with the best *validation* accuracy. Selecting the epoch on the same set you report on is **selection leakage** and inflates every number.

## Data pipeline

- $32 \times 32$  RGB image  $\rightarrow$  **1 024 pixels** as an unordered set
- Each pixel: 3 RGB + 2D Fourier PE = **261 dims** — the same per-pixel width as the paper
- No patching, no convolution, no learned tokenizer

# The Noise Band: How Much Does a Number Move on Its Own?

Three runs, identical in everything but the seed, land 2.78 points apart. Read every other comparison against that.

Three runs, identical in everything but the seed:

| Run                | Seed | Test acc. |
|--------------------|------|-----------|
| e01_baseline       | 42   | 71.63%    |
| e31_baseline_seed1 | 1    | 68.85%    |
| e32_baseline_seed2 | 2    | 70.97%    |
| Spread (max – min) |      | 2.78pp    |

## The rule for every slide that follows

Any difference smaller than **2.78 percentage points** is not an effect. It is seed variance.

## Why this slide exists

Two extra runs cost a few GPU-hours and change the status of all the others: without a measured band, every comparison is an opinion.

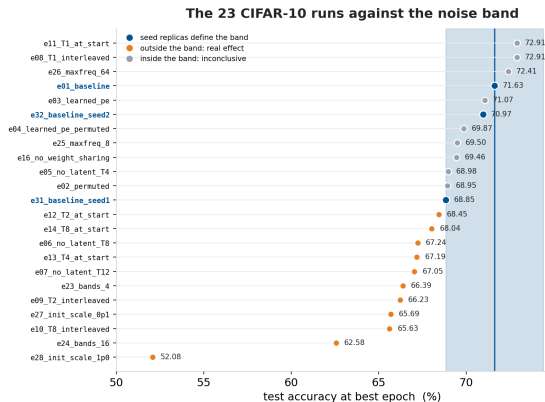
It also sets the honest ceiling of this study. On CIFAR-10 with 1 024 pixels and one GPU, most of the paper's ablations — measured on ImageNet with 50 176 pixels on TPU pods — simply do not have room to show up.

The band depends on the metric: 2.78pp on test accuracy, 2.94pp on best validation, 3.24pp on final validation. The conclusions do not change.

e01 is the *luckiest* of the three seeds. Measured from their mean (70.48%) instead, e12 and e14 fall back inside: 10 runs clear the band, not 12.

# CIFAR-10: What Survives the Noise Band

Ranked against the band: twelve of the twenty-four runs clear it, the other twelve are inconclusive.



24 runs, 12 conclusive

10 against the three-seed mean

- **Outside** the band:  
positional encoding, latent init scale, cross-attend arrangement, Fourier bands, no latent transformer
- **Inside**: permutation, learned PE, weight sharing, max frequency

The chart plots 23 runs: e29\_no\_pe at 32.36% is off scale. Source: each run's results.json, test accuracy at the validation-selected epoch.

# CIFAR-10: Permutation Invariance — the Headline Claim

Shuffling the pixels costs nothing measurable. Removing the encoding costs 39.27 points. That contrast is the claim.

| Comparison                                                          | Positional encoding | Natural → permuted | $\Delta$ |
|---------------------------------------------------------------------|---------------------|--------------------|----------|
| e01 → e02                                                           | Fourier             | 71.63% → 68.95%    | −2.68    |
| e03 → e04                                                           | learned             | 71.07% → 69.87%    | −1.20    |
| e01 → e29                                                           | none at all         | 71.63% → 32.36%    | −39.27   |
| <i>Paper (ImageNet):</i> Fourier 78.0 → 78.0    learned 70.9 → 70.9 |                     |                    |          |

## What we measured

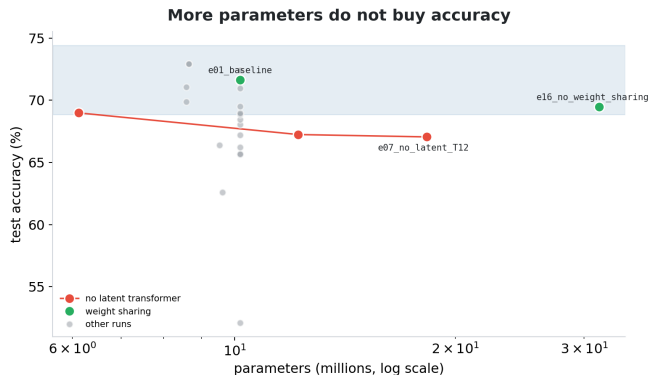
Both drops fall **inside** the 2.78pp band: shuffling the pixels causes no measurable damage. Invariance confirmed — in the honest, weak form: *we did not measure a loss*, which is not the same as proving the loss is zero.

## The control that makes it a claim

Shuffling the pixels costs nothing measurable; **removing the encoding costs 39.27pp**. The same architecture that ignores the order collapses without the coordinates. Position enters through the **encoding**, never through the grid — and that contrast, not either number alone, is the evidence.

# CIFAR-10: Capacity Does Not Buy Accuracy

Three times the parameters do not buy a point: capacity is not what limits this model at CIFAR-10 scale.



## Weight sharing is free

- e01 71.63% with 10.2M params
- e16 69.46% with **31.5M** params
- $3.1\times$  the weights, **-2.17pp** — inside the band

## Where you put capacity matters

- Remove the latent transformer, add cross-attends instead:
- 6.1M  $\rightarrow$  68.98%
- 12.2M  $\rightarrow$  67.24%
- 18.3M  $\rightarrow$  67.05%
- **More parameters, less accuracy**

# CIFAR-10: How Many Cross-Attends, and Where

| $T$ | interleaved              | all at start  | $\Delta$ (arrangement) |
|-----|--------------------------|---------------|------------------------|
| 1   | <b>72.91%</b>            | <b>72.91%</b> | 0.00                   |
| 2   | 66.23%                   | 68.45%        | +2.22                  |
| 4   | <b>71.63%</b> (baseline) | 67.19%        | −4.44                  |
| 8   | 65.63%                   | 68.04%        | +2.41                  |

## The one conclusive result

At  $T = 4$ , interleaving beats front-loading by **4.44pp** — outside the band, even measured from the three-seed mean (3.29pp). Re-reading the input *after* the latents have been refined lets each pass look for something different.

Across rows, reads and depth change together: in our code each read brings its own 4 latent blocks ( $T = 1$ : 4 blocks,  $T = 8$ : 32), while the paper keeps 48 fixed. Within a row the comparison is clean.

## A reproducibility check, for free

At  $T = 1$  the two arrangements produce the *same computational graph*, and indeed give bit-identical results: 72.91%, epoch 90, 8 654 662 params. Fixed seed  $\Rightarrow$  reproducible training.

# CIFAR-10: The Hyperparameter That Actually Matters

| Sweep                | Value       | Test acc.     |
|----------------------|-------------|---------------|
| Latent init $\sigma$ | <b>0.02</b> | <b>71.63%</b> |
|                      | 0.1         | 65.69%        |
|                      | 1.0         | 52.08%        |
| Fourier bands        | 4           | 66.39%        |
|                      | 16          | 62.58%        |
|                      | <b>64</b>   | <b>71.63%</b> |
| Max frequency        | 8           | 69.50%        |
|                      | <b>16</b>   | <b>71.63%</b> |
|                      | 64          | 72.41%        |

## Latent initialisation: the only clean monotone effect

$\sigma$  from 0.02 to 1.0 costs **19.55pp**. At  $\sigma = 1.0$  the best epoch is **9 out of 120**: the model learns briefly, then degrades for the remaining 111 epochs.

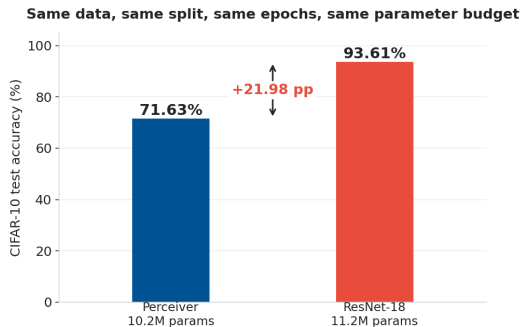
## Two honest caveats

**Bands** is non-monotone (4 beats 16), and both low points are runs whose *final* validation collapsed (22.5% and 55.5%) — an optimisation problem, not a resolution one.

**Max frequency** sits entirely inside the noise band. The whole axis is inconclusive, and saying so is part of the result.

# CIFAR-10: the Reference That Is Not a Perceiver

Neither Perceiver paper benchmarks CIFAR-10, so we trained the reference ourselves: a ResNet-18, same split, same epochs, same budget.



## The uncomfortable comparison comes first

A **ResNet-18** on the same split, the same 120 epochs and 11.2M parameters against 10.2M reaches **93.61%**, where e01\_baseline reaches **71.63%**: **-21.98pp**.

Neither Perceiver paper benchmarks CIFAR-10, so without this run there is nothing to read the number against.

## What the gap actually measures

On ImageNet the paper *beats* ResNet-50 by +4.5pp (78.0 vs 73.5). What differs between the two settings is not the architecture, it is the **scale**.

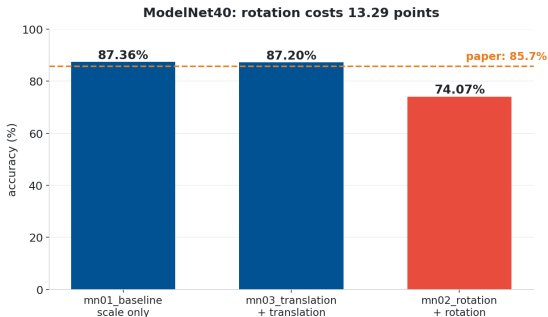
The Perceiver's claim was never accuracy — it is **having no prior on the domain**. Here that prior is worth 22 points.

# ModelNet40: Same Network, 3D Point Clouds

The same network, fed 2,048 points in 3D instead of pixels. Only the positional encoding changes dimension.

| Setting               | Value                |
|-----------------------|----------------------|
| Latent array          | $128 \times 512$     |
| Cross-attn / blocks   | 2 / 6                |
| Points per model      | 2 048                |
| Input per point       | 42 dims (3 + 39)     |
| Fourier bands / max f | 6 / 1120 (paper: 64) |
| Optimizer / lr        | LAMB / 0.001         |
| Batch (paper: 512)    | 32                   |
| Epochs                | 120                  |
| Parameters            | 23 288 928           |

**Metric note:** ModelNet40 has no separate validation split — the standard test set (2 468 objects) is the validation set, so the figure is `val_accuracy` at the best epoch. The last epoch, with no selection at all, gives **86.43%**.



## Above the paper

**87.36%** vs the paper's **85.7%**, with a batch  $16\times$  smaller — and still above at the last epoch. Plausible: ModelNet40 is small and its objects arrive canonically aligned, so the large-batch advantage that dominates ImageNet counts for little here.

# ModelNet40: Rotation Costs 13 Points — and Why

Rotation costs 13.29 points — the positional encoding again. Scale and translation never reach the network.

| Augmentation   | Accuracy      | $\Delta$      |
|----------------|---------------|---------------|
| scale only*    | <b>87.36%</b> | —             |
| + translation* | 87.20%        | −0.16         |
| + rotation     | <b>74.07%</b> | <b>−13.29</b> |

## \* Two of the three runs see the same data

The normalisation re-centres each cloud and rescales it to unit radius, so a scale or a shift of the *whole* cloud is undone before the model sees it — the paper jitters *each point* instead. `mn01` and `mn03` are therefore a free replica: 0.16pp is seed noise.

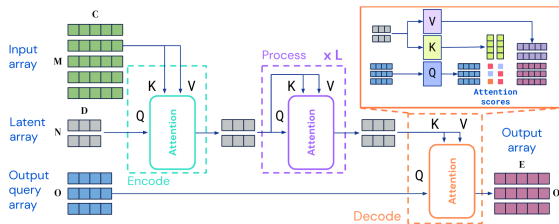
## The Perceiver has no rotational inductive bias

3D Fourier PE encodes **absolute** position. Rotating an object changes every coordinate — and therefore every encoding — while the label stays the same. The network must learn from scratch an invariance the architecture does not grant, and that this task does not even need: ModelNet40 objects are already canonically aligned.

Here the axis is arbitrary in 3D and the rotation is applied in training only: the test objects stay upright. The paper also found that rotation did not help.

**Takeaway:** a missing inductive bias cannot be patched with augmentation.

# Perceiver IO: Structured Outputs



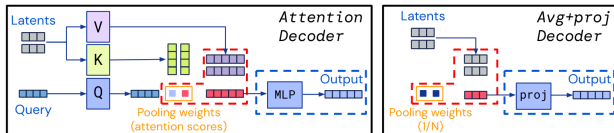
**Extension over Perceiver** (Jaegle et al., ICLR 2022):

- **Encode:** input  $\rightarrow$  latent (same cross-attn encoder)
- **Process:** latent self-attention ( $\times L$ )
- **Decode:** output queries read latents

**Why it matters:**

- Original Perceiver: one pooled global output
- IO: task-specific output shape
- Same latent bottleneck, more general decoder

# Perceiver IO Decoder with Output Queries



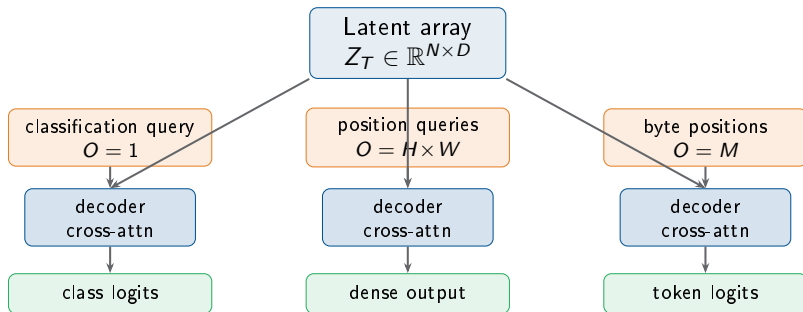
## Decoder cross-attention (single layer):

$$Q_{\text{dec}} = \text{OutputQuery } W_Q, \quad K_{\text{dec}} = \text{Latent } W_K, \quad V_{\text{dec}} = \text{Latent } W_V$$

$$\text{Output} = \text{softmax}\left(\frac{Q_{\text{dec}} K_{\text{dec}}^T}{\sqrt{d_k}}\right) V_{\text{dec}} \in \mathbb{R}^{O \times D}$$

- A task head then projects each of the  $O$  rows to the output channels  $E$  (logits, pixels, bytes)
- The number of output queries  $O$  is **task-dependent** and independent of  $N$  and  $M$
- A single decoder cross-attention layer suffices (no self-attention among outputs)

# Output Query Design per Task



## Classification

One learned query produces one global prediction.

## Dense outputs

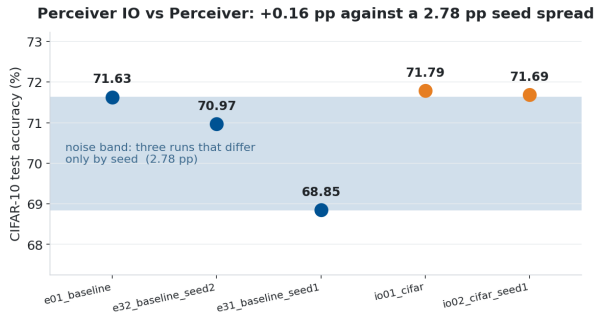
Output shape is controlled by the query array.

## Masked LM

One query per byte; the loss counts only the masked ones.

# Perceiver IO on CIFAR-10: the Decoder Changes Nothing Here

On a single label out of ten, the query decoder and mean pooling are the same thing — and the measurement says so.



## What changed, and what did not

Same encoder as e01\_baseline; the mean-pooling head is replaced by a **decoder with one output query**, and the training recipe follows the IO paper (App. A.1):  $\text{lr } 10^{-3}$ , cosine.

io01\_cifar **71.79%** vs e01 **71.63%**: +0.16pp, against a seed spread of **2.78pp**. The replica io02 lands at 71.69%.

## This was the expected outcome

No difference is the result, not a disappointment. The decoder earns its keep when the output is **structured** — one query per byte in the MLM, one per task in the multitask run. For one label out of ten, querying the latents and averaging them are the same thing.

# WikiText-103: Byte-Level Masked Language Model

Pre-train a language encoder with no tokenizer at all: the input is the raw UTF-8 byte stream.

| Setting                        | Value               |
|--------------------------------|---------------------|
| Latent array                   | $128 \times 512$    |
| Cross-attn / blocks            | 1 / 4               |
| Heads (cross / self)           | 1 / 8               |
| Sequence length                | 512 bytes           |
| Vocabulary                     | $256 + 1$ mask      |
| Masking                        | 15% of single bytes |
| Output queries                 | one per position    |
| Optimizer / lr                 | LAMB / 0.001        |
| Epochs / batch                 | 10 / 32             |
| Parameters                     | 18 872 740          |
| <b>Masked-byte accuracy</b>    | <b>86.68%</b>       |
| Lookup of the two neighbours   | 42.96%              |
| Always a space (most frequent) | 19.07%              |
| Uniform chance (1/256)         | 0.39%               |

## Where the decoder finally earns its keep

Nearly nine masked bytes out of ten, reading raw UTF-8 with **no tokenizer at all** — **44 points above** a lookup table of the two neighbouring bytes. That gap is what the long context buys.

## Why bytes are affordable here

Cross-attention costs  $O(M \cdot N)$  — *linear* in sequence length — instead of  $O(M^2)$ . Going from 512 tokens to 2 048 bytes costs BERT 16 $\times$  in attention, and Perceiver IO only 4 $\times$ .

## Read this number carefully

We mask *single bytes*, the paper *whole words*: a masked byte usually keeps its neighbours, which is why the lookup already reaches 42.96% (`baseline_mlm.py`). Nor is it comparable to masked-*token* accuracy — the comparison with BERT lives on GLUE.

# GLUE: Eight Tasks, One Encoder, No Tokenizer

The same encoder, the same weights, one query swapped — and two of the eight numbers turn out not to be learning.

| Run                        | Task              | Budget | Result       |
|----------------------------|-------------------|--------|--------------|
| io_glue_colab              | acceptability     | 30     | 69.13        |
| io_glue_sst2               | sentiment         | 10     | 80.73        |
| io_glue_mrpb               | paraphrase        | 30     | 70.59        |
| io_glue_stsb               | similarity (reg.) | 30     | 60.82        |
| io_glue_qqp                | duplicate Q       | 3      | <b>84.66</b> |
| io_glue_mnli               | NLI (3-way)       | 3      | 65.80        |
| io_glue_qnli               | QA entailment     | 10     | 80.65        |
| io_glue_rte                | entailment        | 30     | 56.68        |
| Mean over the eight        |                   |        | <b>71.13</b> |
| Paper, Tab. 1 (byte-level) |                   |        | 81.0         |

STS-B is Pearson  $\times 100$ , the official GLUE metric. Grey marks the two majority-class predictors. Fine-tuning starts from the MLM checkpoint; the epoch budget follows task size (393 k examples against 2.5 k).

## Two of these numbers are not learning

**CoLA** scores 721 correct out of 1 043 — *exactly* the majority-class count. **MRPC** beats its majority by 9 examples out of 408. Both are constant predictors wearing an accuracy.

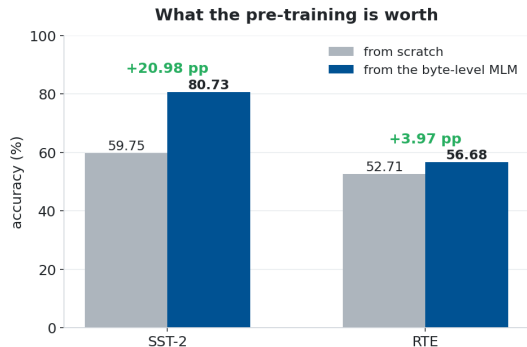
The mean of 71.13 therefore rests on **six** tasks, not eight. CoLA in the paper is scored with Matthews' correlation, which for a constant predictor is **0**.

## The gap that does not close

71.13 against the paper's 81.0. The scale is not comparable: **18.9M** parameters pre-trained on WikiText-103 for 10 epochs, against **201M** on Wikipedia + C4 for 500 000 steps.

# GLUE: What the Pre-training Is Actually Worth

Two tasks fine-tuned twice — once from the MLM checkpoint, once from random weights. Everything else identical.



## Why two controls and not eight

A fine-tuned 80% means nothing on its own — maybe the task is solvable at 80% from random weights. The two scratch runs are chosen at the **extremes** of data size: **SST-2** has 67 000 training examples, **RTE** has 2 500. Everything else is identical.

## And the prediction holds

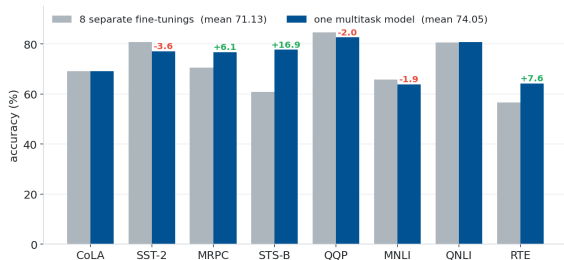
SST-2: 59.75% → 80.73%, +20.98pp. RTE: 52.71% → 56.68%, +3.97pp.

Transfer is worth five times more where the data abounds. The small task cannot exploit the encoder it was given: 2 500 examples are not enough to move it far from its starting point.

# GLUE Multitask: the Comparison That Replicates Best

One model with eight output queries against eight separate fine-tunings — the comparison from Table 2 of the IO paper.

One model with eight output queries beats eight models: 74.05 vs 71.13



## Table 2 of the IO paper

One model with **one output query per task**, against eight separate fine-tunings. Paper: 81.8 vs 81.0, +0.8. Here: 74.05 vs 71.13, **+2.92**.

The absolute level stays ~8 points below. But **sign and order of magnitude hold** — and that relation, not the level, is what Tab. 2 claims.

## Where the gain sits

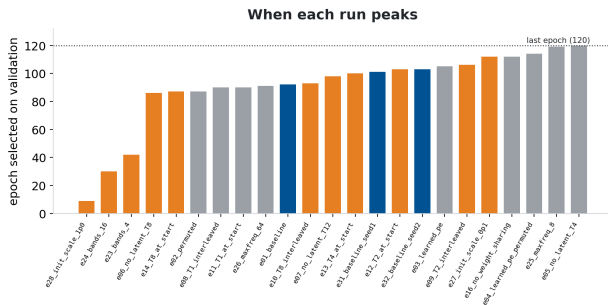
All on the small tasks — STS-B +16.9, RTE +7.6, MRPC +6.1 — which see only 1.35 passes over their own data. The large ones lose 2 to 3.6 points; CoLA stays at the majority class in both.

## Not a fair fight — in our favour

The multitask model selects **one** epoch for all eight tasks; the separate models select **eight**. It wins anyway — and a ninth task would cost **one query**, not another model.

# Convergence: When Each Run Peaks

There is no early stopping: every run trains the full schedule and reports the checkpoint that was best on validation.



## Why this matters

The distribution of best epochs is itself a diagnostic: a run that peaks early and then decays is unstable, whatever its best number says.

## Three unstable runs

Validation accuracy, best epoch → final epoch:

e23: 67.74% @42 → **22.54%**

e24: 62.66% @30 → **55.46%**

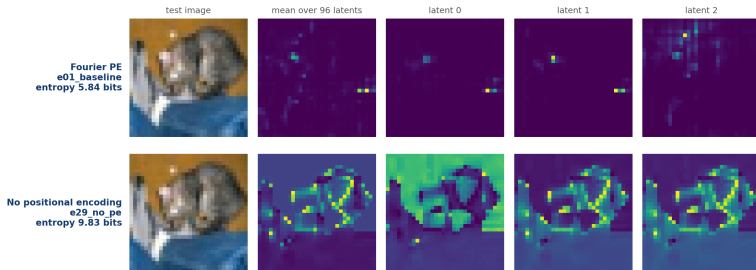
e28: 53.36% @9 → 46.32%

Their reported numbers come from the validation-selected checkpoint — the correct methodology — but the instability has to be declared, not hidden.

e05\_no\_latent\_T4 peaks at epoch **120**: it was still improving when the schedule ended.

# Attention Maps: Where the Latents Look

With coordinates the latents pick locations; without, they can only pick colours.



## Fourier PE (e01, 71.63%)

Each latent attends to a few pixels at specific locations: mean entropy **5.84 bits** over 1 024 pixels (uniform: 10 bits). Latents specialise by region.

## No PE (e29, 32.36%)

Same-colour pixels are indistinguishable: attention follows colour boundaries and spreads over the image — **9.83 bits**, 98% of uniform.

# Comprehensive Results Summary

Forty-two runs, all completed. Every number below traces to a `results.json` — none is quoted from memory.

| Modality                  | Model        | Best Acc.         | Reference           | Experiments    |
|---------------------------|--------------|-------------------|---------------------|----------------|
| Images (CIFAR-10)         | Perceiver    | 72.91%            | 93.61% <sup>†</sup> | <b>24</b> runs |
| Images (CIFAR-10)         | Perceiver IO | 71.79%            | 71.63% (Perceiver)  | 2              |
| Point Clouds (ModelNet40) | Perceiver    | <b>87.36%</b>     | 85.7%               | 3              |
| Text (WikiText-103)       | Perceiver IO | <b>86.68%</b> MLM | 42.96% lookup       | 1              |
| NLU (GLUE, 8 tasks)       | Perceiver IO | 71.13 mean        | 81.0 (paper)        | 11             |
| Images (CIFAR-10)         | ResNet-18    | <b>93.61%</b>     | —                   | 1              |

<sup>†</sup> A ResNet-18 trained here under identical conditions. Neither Perceiver paper benchmarks CIFAR-10, so the reference had to be measured rather than quoted.

## What is measured

- ModelNet40 **beats** the paper: +1.66pp (+0.73 at the last epoch)
- Byte-level MLM: 86.68%, 44pp above a neighbour lookup
- One multitask model beats eight separate ones: 74.05 vs 71.13
- Noise band **2.78pp**, measured on 3 seed replicas
- 12 of 24 CIFAR runs are conclusive (10 against the seed mean)

## Scope of the campaign

- Declarative registry of **42 runs, 42** completed
- 3 modalities: 2D images, 3D point clouds, raw bytes
- One command per run: `experiments.py -run <id>`
- Every figure above traces to a `results.json`; none is quoted from memory

# Gap vs Original Papers

On the modality with the weakest domain prior we beat the paper. On the one with the strongest, a convolutional net of the same size beats us.

## ModelNet40 (Point Clouds):

| Source              | Acc.           |
|---------------------|----------------|
| Original Paper      | 85.7%          |
| This implementation | <b>87.36%</b>  |
| Gap                 | <b>+1.66pp</b> |

### We are above the paper.

With a batch  $16\times$  smaller (32 vs 512). Plausible because ModelNet40 is small and canonically aligned, so the large-batch advantage counts for little. The epoch is picked on the test set; the last one, 86.43%, is still above.

### What the two columns say together

On the modality with a **weak** domain prior — 3D point clouds — we are above the paper. On the one with a **strong** prior, a convolutional net of the same size wins by 22 points. On ImageNet the paper beats ResNet-50 by +4.5pp: the prior stops paying at scale, and **that is the claim being replicated**, not an absolute accuracy.

## CIFAR-10 (Images):

| Source                                 | Acc.                    |
|----------------------------------------|-------------------------|
| ResNet-18, same budget<br>e01_baseline | <b>93.61%</b><br>71.63% |
| Gap                                    | <b>−21.98pp</b>         |

### The reference had to be measured.

Neither Perceiver paper benchmarks CIFAR-10, so a convolutional net was trained here under identical conditions rather than quoting a number from elsewhere.

# Hardware Limitations & Design Choices

Every limitation below is one GPU instead of a TPU pod. The mechanism is the paper's; the budget is not.

| Parameter           | Original Paper (Google)           | Project Setup                |
|---------------------|-----------------------------------|------------------------------|
| Hardware            | TPU pods (up to 64 TPUs)          | 1 NVIDIA RTX 3080            |
| Batch size          | up to 1024                        | 32–64                        |
| Latents ( $N$ )     | 256–512                           | 96–128                       |
| Model size          | 44.9M (Perceiver), 201M (IO text) | 10.2–23.3M params            |
| Pre-training corpus | Wikipedia + C4, 201M-param IO     | WikiText-103, 18.9M-param IO |

## Consequences:

- LAMB at batch 32–64, not the 512–1024 it targets
- 96–128 latents instead of 512
- One seed per ablation, three for the baseline
- No search beyond the registered ablations

## Mitigation strategies:

- Weight sharing: 10.2M parameters, not 31.5M
- bf16 autocast + gradient clipping (LAMB at lr 0.004)
- A measured noise band, not more runs per cell
- Validation carved from train; test touched once

# Possible Improvements

## Architecture:

- Increase latent count ( $N = 256+$ ) with gradient checkpointing
- Add multi-scale cross-attention (coarse  $\rightarrow$  fine)
- Explore relative positional encodings (RoPE, ALiBi)
- Deeper decoder for structured outputs

## Training:

- Larger batch sizes with gradient accumulation
- More seeds per ablation, to narrow the 2.78pp band
- Longer training schedules
- Advanced augmentation (CutMix, MixUp)

## Data & Tasks:

- ImageNet pre-training for vision
- Longer byte-level pre-training: more epochs, larger corpus
- Multi-modal joint training
- Audio modality experiments

## Analysis:

- Layer-wise attention probing
- Latent space clustering visualization
- Systematic hyperparameter search
- Comparison with efficient Transformers (Linformer, Performer)

Everything measured, split by whether it survived the noise band.

## Conclusive (outside the 2.78pp band):

- **Positional encoding** is the whole thing: removing it costs **39.27pp** (71.63→32.36%)
- **Generality**: one architecture, 87.36% on 3D point clouds — above the paper — and 86.68% on raw bytes
- **Pre-training transfers**: SST-2 59.75→80.73%, and 5× less on the small task
- **Latent init scale** is the critical hyperparameter: 0.02→71.63%, 1.0→52.08%
- **Interleaving** cross-attends beats front-loading by 4.44pp
- **Rotation augmentation** costs 13.29pp on ModelNet40

## Inside the band — “no measurable damage”:

- Permutation invariance, with *both* positional encodings
- Weight sharing: 3.1× fewer parameters at no measurable cost
- Perceiver IO vs Perceiver on images: +0.16pp
- ModelNet40 translation: −0.16pp — undone by the normalisation, a free replica

## Measured, and not in our favour:

- A ResNet-18 on the same budget: **93.61%** vs 71.63%
- GLUE mean 71.13 against the paper's 81.0
- Two GLUE tasks (CoLA, MRPC) are majority-class predictors
- No configuration beats the baseline outside the band — every conclusive effect is **negative**

## Central methodological finding

Measuring the **noise band** — three replicas differing only by seed — cost two runs and changed the status of all the others. Without it, 12 of the 24 comparisons (14 against the seed mean) would have been reported as effects when they are variance.

## What the architecture gives, and what it still costs at this scale.

### Architecture properties:

- Single architecture for images, point clouds, and text
- Linear complexity in input size
- Weight sharing:  $3.1\times$  fewer parameters at no measurable cost
- Zero CNN/RNN domain-specific assumptions
- Output queries: one more task costs **one query**, not another model

### Remaining limitations:

- At this scale the domain prior is worth **22 points** (93.61 vs 71.63)
- Entirely dependent on the positional encoding (−39.27pp without it)
- GLUE sits  $\sim 10$  points below the paper (71.1 vs 81.0), at 1/10 of the parameters and pre-training
- LAMB at batch 64 is sensitive to the latent init scale
- Half the image ablations are indistinguishable from noise

### Closing statement

A **single general-purpose architecture** does apply to images, point clouds and raw bytes — and on the modality with the weakest domain prior it beats the paper. Its advantage was never accuracy: it is **having no prior on the domain**. At CIFAR-10 scale that prior is worth 22 points; the paper's own claim is that it stops paying at ImageNet scale, and nothing measured here contradicts it.

# Thank you!

Questions?

Francesco Gigli and Corso Vignoli

Code and reproduction instructions shared with the course